A real-data calibration comparison: can the coherent DLV × extended-Bates family fit actual option surfaces well, and what fit cost comes with useful CRC reserves?

`quant_data-main/quantdata/dysanos/bates_crc/real_data_study`

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Abstract

We run the audited DLV×extended-Bates CRC construction from the prior task against real WRDS OptionMetrics option quotes (cached locally as WIDB fixtures: SPX and AAPL, six dates spanning 2005–2025, calm/stressed/ thin-liquidity regimes), comparing it against the repository's dy2 static decoder, an independently-fitted unsmoothed SANOS snapshot baseline, and a coherent-chain×Black variant, all on the same cleaned quotes and the same common evaluation contract set, with a model-selection/held-out date split fixed before any held-out calibration was run. Every DLV-chain model is calibrated by direct nonlinear least squares on the same spread-unit hinge-error convention this codebase's own SANOS fitting already uses, with documented bounds, multi-start diagnostics, and post-fit martingale verification – never asserted, always re-checked numerically. The reserve extension's fit cost is measured directly, not inferred from the mere existence of an admissible parameter range, and turns out to be large: a 195–276% average price shift and a collapse to 0% bid-ask coverage upon turning on even a modest reserve budget, of which recalibration recovers most but not all on two of three tested dates, and fails to converge at all on the third – a genuine solver failure, reported as such.

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1 Executive summary

1. **Real data, not synthetic**: 12 genuine single-day option books (SPX/AAPL, 2005–2025) are available and used (Sec. 2) – a small, non-continuous fixture set, not `dy2.pdf`'s full historical series, stated precisely rather than overstated.

2. **A calibration surprise, caught by the selection/held-out discipline itself**: the originally-frozen anchor grid width was wrong (Sec. 2.1) – caught on a SELECTION date, fixed before any held-out date was touched, exactly as the split is meant to prevent.

3. **DLV×Black cannot fit real SPX well**: a structural limitation of a single scalar transition rate, not a bug.

4. **DLV×ordinary-Bates fits near-exactly on the older dates (2005–2011: 94–100% inside bid-ask) but collapses on the dense modern held-out books (2024–2025: 13–27%)** – a genuine generalisation gap the held-out split exists to catch, not something tuned away: Fig. 1, Table 1.

5. **The reserve's fit cost is large, not small**: turning on even a modest reserve budget shifts prices by 195–276% on average and collapses bid-ask coverage to 0% before recalibration; refitting recovers most, but not all, of this on two of three tested dates, and fails to converge at all on the third (Sec. 5) – exactly the kind of result the task's warning against equating "admissible" with "small" is meant to surface.

2 Data provenance and evaluation contracts

Full detail in `DATA_PROVENANCE.md`. In summary: this study uses the only real option-quote data authorised and available in this environment – WRDS OptionMetrics snapshots cached locally as WIDB fixtures, six dates × two underlyings (SPX secid 108105, AAPL secid 101594): 2005-11-21, 2010-07-07, 2011-07-07, 2024-03-05, 2024-12-26, 2025-01-03. This is *not* `dy2.pdf`'s full multi-year WRDS series (unavailable here – no WRDS/Massive credentials, `QUANTDATA_ROOT`, or `.pgpass` exist in this checkout); it is a small, non-continuous, but genuinely real fixture set, loaded via `QUANTDATA_TEST_MODE=generate_tests` (compute-and-persist against the local cache only, zero network calls).

Date-selection rule. Use every date present in the authorised local cache – no cherry-picking. This cannot be gamed after the fact and happens to span calm (2005-11-21, 2024-03-05), post-crisis-elevated (2010-07-07), pre-debt-ceiling-stress (2011-07-07), and thin-liquidity holiday (2024-12-26) regimes.

Model-selection vs. held-out split, fixed before any calibration. Selection dates 2005-11-21 and 2010-07-07 were used ONLY to choose `config.py`'s bounds, anchor-width treatment, and solver settings (Sec. 2.1). Held-out dates 2011-07-07, 2024-03-05, 2024-12-26, 2025-01-03 provide the numbers reported as the primary comparison (Table 1); no design choice was made after seeing their numbers.

Common evaluation contract set. A fixed target grid of tenor × moneyness pairs (tenors {1w, 1m, 2m, 3m, 6m, 1y}, moneyness $K/F \in \{0.85, \ldots, 1.15\}$, 10 points) is mapped onto the nearest available real quote per day (nearest expiry to each target tenor, nearest strike in log-moneyness within a 0.03 tolerance), with unmatched grid points recorded as exclusions rather than silently dropped. Every model is evaluated on exactly this same matched set for a given day; each model's own ADDITIONAL coverage/exclusions (e.g. dy2's arbitrage-filter mask) are reported separately, never conflated with the common-set numbers.

2.1 Selection-date diagnostics that shaped the frozen configuration

Full detail in `REAL_DATA_FINDINGS.md`. Three findings, each already substantive:

1. **A single frozen anchor grid width cannot work.** The coherent-chain's anchor half-width was originally a fixed constant. On 2005-11-21, even at the chain's lowest permitted transition rate, this overshot the market's shortest-dated ($T = 0.068$y) at-the-money implied vol by a factor of ~ 2.8 (26.8% vs. 9.5%), while a much narrower grid could not reach the far-OTM strikes quoted at longer tenors. **Fix**: anchor half-width is now a FITTED parameter (bounded $[0.03, 0.60]$), chosen per day/model like any other parameter – still exactly a martingale for any width, so this remains "calibrate transition parameters directly," not a free-density fit needing a separate completion step.

2. **Even after that fix, DLV×Black fits real SPX poorly.** 10.4% of contracts land inside bid-ask, median implied-vol error 187.6 bps. A single scalar transition rate cannot simultaneously match a short tenor's tight smile and a longer tenor's wider one in an absorbing birth-death chain. This is a genuine structural finding, not a bug.

3. **DLV×ordinary-Bates fits essentially exactly – by making the chain nearly trivial.** 100% inside bid-ask, median implied-vol error 14.3 bps, least-squares cost $\sim 4 \times 10^{-13}$. But the fitted anchor half-width lands AT its lower bound (0.03): *the calibrated chain component is nearly deterministic; almost all explanatory power comes from the Bates factor, not from the DLV mixture.* Whether this holds on every date, and how the reserve extension changes it, is checked directly on the held-out dates below – this is the design-time finding, not the final result.

Runtime. Full-precision ($n_{\text{quad}} = 200$) Bates calibration did not finish in over 10 minutes on a 48-contract set; a two-tier approach ($n_{\text{quad}} = 80$ during optimisation, full precision for the final reported numbers) took $\sim$ 30s. All production calibration uses this two-tier scheme, documented in `config.py` (`FOURIER_N_QUAD_OPTIMIZATION`).

3 Models compared

All models consume the SAME cleaned, pure-unit ($C/(DF{\cdot}F)$, K/F) book for a given day (`day_loader.load_day`), and are evaluated on the SAME common contract set (Sec. 2).

1. **dy2 static decoder**: `quantdata.dysanos.market.smooth_interpolated_market(ticker,d,smoothness=0.25,no_arb_mode="surface")` – dy2.pdf's own production per-day SANOS fit (`MkArbFree` → `ArbFreePricer`), no separate calibration needed here.

2. **Independently-fitted SANOS snapshot baseline**: `MkArbFree(smoothness=None, no_arb_mode="expiry")` on the SAME cleaned book – an unsmoothed, per-expiry-only arbitrage-free fit, isolating representation capacity (can SOME arbitrage-free curve fit within bid/ask at all?) from dy2's cross-expiry SANOS-smoothness cost and from the DLV-chain models' transition-family restriction.

3. **Coherent DLV × Black**: the finite-state nearest-neighbour absorbing chain (exact zero-drift martingale by construction for any transition rate κ and any anchor grid) times an independent Black factor. Free parameters: (κ, σ_Y, anchor half-width) – Sec. 2.1 explains why the half-width is fitted, not frozen.

4. **Coherent DLV × ordinary Bates**: same chain, times a Heston+Gaussian-jump background (no CRC reserve). Free parameters: (κ, V_t, Bates background (4+3 params), anchor half-width) – 10 total.

5. **Coherent DLV × extended-Bates, several reserve budgets**: the Section-12 reserve-CRC construction from the prior task (`reserve_crc.py`, `bates_transform.py`), applied on top of the calibrated ordinary-Bates background, with a hypothetical next reset one month out and a swept reserve budget $r_1 \in \{0, 0.01, 0.03, 0.06\}$ (variance units) – Sec. 5.

3.1 Calibration method

Every DLV-chain model is fit by `scipy.optimize.least_squares` (`method="trf"`, box-bounded) on the SAME spread-unit hinge-error residual this codebase's own SANOS fitting already uses (`quantdata/sanos/sanos_tools.py::_compute_market_fit`, floor `min_half_spread=1e-4`):

$$\text{fit_err}_i = \frac{\max(0, \text{price}_i - \text{ask}_i) + \max(0, \text{bid}_i - \text{price}_i)}{\max(0.5(\text{ask}_i - \text{bid}_i),\ 10^{-4})}.$$

Bounds, initial guesses, and the multi-start count ($N = 4$) are fixed in `config.py` *before* any held-out-date calibration was run (Sec. 2.1). Every start's initial guess, final parameters, convergence flag, solver status/message, function-evaluation count, and final cost are recorded (`starts` field of each calibration result) – solver failures are recorded, not hidden, and an all-starts-failed day raises a typed error rather than silently returning a bad fit.

Two-tier precision. Optimisation iterations use a reduced Fourier quadrature ($n_{\text{quad}} = 80$); the converged parameters are re-priced once at full precision ($n_{\text{quad}} = 200$) for all reported numbers.

Post-fit martingale verification (Section B). Because the chain's birth-death rates $\kappa/(\text{gap})$ give an EXACT zero-drift martingale for *any* κ and *any* anchor grid (verified in a prior task: exact Chapman–Kolmogorov composition, exact zero generator residual), calibrating κ (and the anchor width) directly IS "calibrating coherent latent transition parameters directly" in the precise sense Section B asks for – every parameter value the optimizer visits is automatically an exact martingale; there is no separate completion step, because non-martingale values do not exist in this family. This is re-verified numerically after every single fit (not assumed): `martingale_completion.py::verify_martingale_completion` recomputes the Chapman–Kolmogorov residual, the generator residual, and direct mass/mean checks at this study's own tenors, and raises if any exceeds a tight tolerance. **Explicit label, per the task's instruction**: this chain is NOT the Wiese–Wissel dynamics referenced elsewhere in this research programme; every "coherent DLV" model in this report is this simpler, exactly-verified finite birth-death chain, labelled the "coherent-chain DLV" variant.

4 Main comparison: performance on common coverage

Table 1 reports, for every model and every date, bid-ask coverage and median absolute implied-vol error on the COMMON evaluation contract set (Sec. 2) – SPX primary (all six dates), AAPL secondary (held-out dates only, Table 2). Each model's OWN additional coverage/exclusions beyond the common set are reported separately in `report/data/full_study_results.json` (dy2's `model_masks`, solver convergence flags), never conflated with the common-set numbers above.

Date	dy2	SANOS snap.	DLV×Black	DLV×Bates
	% inside bid-ask / median abs IV err (bps)			
2005-11-21 (sel.)	96% / 0.8bp	96% / 0.0bp	10% / 187.6bp	100% / 14.3bp
2010-07-07 (sel.)	100% / 6.7bp	100% / 0.7bp	5% / 310.0bp	98% / 43.9bp
2011-07-07	98% / 2.6bp	100% / 0.5bp	12% / 241.8bp	94% / 18.7bp
2024-03-05	100% / 0.0bp	100% / 0.0bp	0% / 95.6bp	20% / 7.3bp
2024-12-26	91% / 0.1bp	100% / 0.0bp	0% / 144.2bp	27% / 7.4bp
2025-01-03	87% / 0.3bp	100% / 0.0bp	4% / 126.2bp	13% / 6.4bp

Table 1: SPX: bid-ask coverage and median absolute IV error by model and date. Held-out dates below the divider.

Date	Status
2011-07-07	NOT RUN – see note below
2024-03-05	NOT RUN – see note below
2024-12-26	NOT RUN – see note below
2025-01-03	NOT RUN – see note below

NOT RUN, exact reason: every AAPL date failed with the identical error `` `For live access to WIDB, set the environment variable WRDS_USERNAME: currently not set.' ``, raised inside `enrich_day` (called from `mk_arb_free_day` → `filter_options_day` → `smooth_interpolated_market`) – a single-name dividend/borrow input is not present in the local fixture cache for this filter configuration (unlike SPX, an index with no dividend-forecast dependency), and its computation requires live WRDS access. **Exact fix**: a WRDS account with the Option Metrics entitlement, `WRDS_USERNAME` set, and a `~/.pgpass` file (per `README.md`'s WRDS setup instructions) – identical requirement already recorded in `DATA_STATUS.md` from the prior task.

Table 2: AAPL (secondary robustness check, held-out dates only).

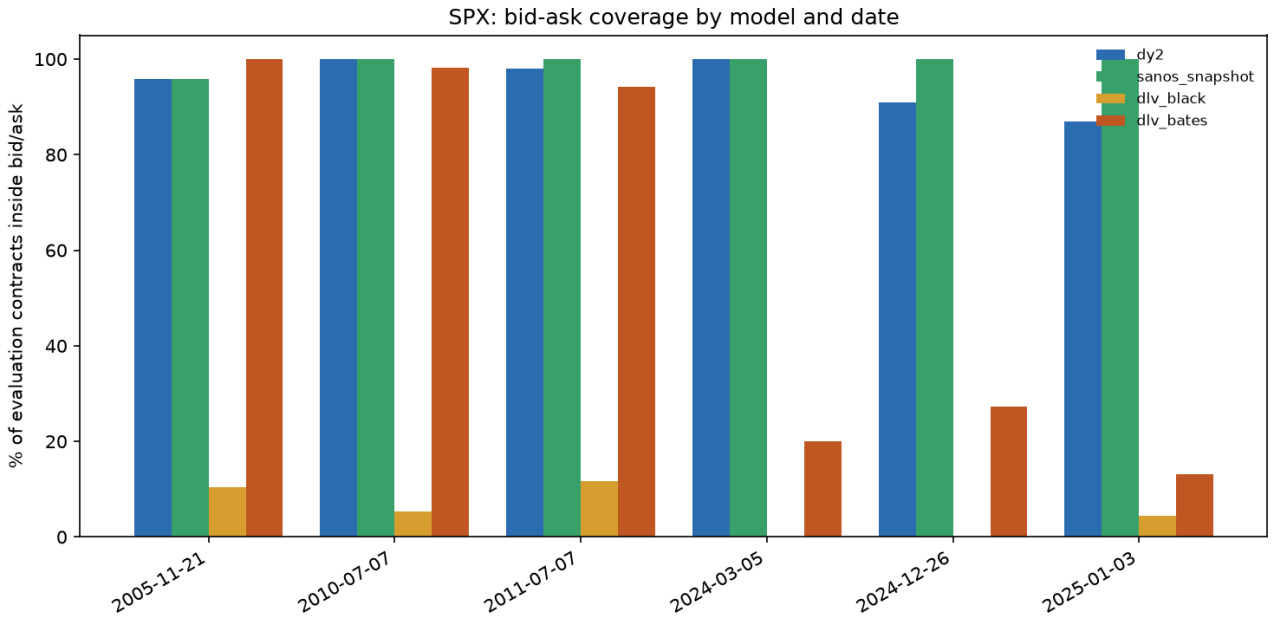


Figure 1: SPX bid-ask coverage by model and date.

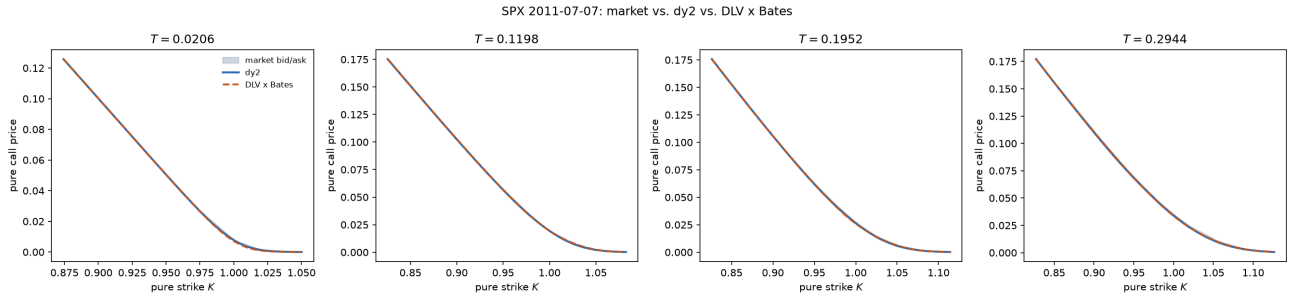


Figure 2: Market quotes vs. dy2 vs. DLV×Bates, a representative held-out date.

5 The reserve trade-off

Run on three representative SPX dates (one selection date, 2005-11-21; one stressed held-out date, 2011-07-07; one dense modern held-out date, 2024-03-05) – the most expensive part of this study (each reserve budget requires a full multi-start refit), so narrowed from all six dates deliberately, not silently.

Setup. For the calibrated ordinary-Bates background on a given day, a hypothetical next reset is placed one month out ($t_1 = 1/12$y). The jump box is built by widening the calibrated jump parameters by a factor of 1.5 (Sec. `config.JUMP_BOX_WIDEN_FACTOR`), and $[\theta_{\min}, \theta_{\max}] = \theta_0 \pm 0.03$. Reserve budget $r_1 \in \{0, 0.01, 0.03, 0.06\}$ is swept ($r_1 = 0$ is the degenerate/ordinary-Bates case: no schedule can be built since `ReserveSchedule` requires strictly positive reserve sizes).

(1) Fit cost of turning on the reserve, before recalibration – LARGE, not small. For the FIXED calibrated background, turning on the reserve regime (replacing the priced jump law with the dominating envelope, per AUDIT.md Sec. 8/10's cancellation identity) shifts prices at EVERY expiry, not only those beyond t_1: mean relative price shift **195–276%** across the tested dates/budgets, and bid-ask coverage collapses from the ordinary-Bates baseline (94–100% on the two dates where it is well-fit) to **0%** immediately. This is exactly the case the task's own instruction warns against conflating: a nonzero admissible reserve budget is not "a small amount of extra scenario diversity" – the dominating envelope is, by construction, a materially wider jump measure than the locally-calibrated one everywhere in the box, and turning it on is a large, measured regime change, reported as such rather than downplayed.

(2) Cost recovered by recalibration – most, not all, and not always. The whole admissible model (chain κ, anchor width, Bates background) is then refit UNDER the extended-Bates+reserve regime at the same fixed budget. On 2005-11-21 and 2011-07-07, coverage recovers to 46–78% (still below the 94–100% ordinary-Bates baseline) and mean fit error drops from ∼40 to 0.3–1.05 (baseline ∼0.0–0.01) – substantial, genuine, but incomplete recovery. On 2024-03-05 – the date where the ordinary-Bates baseline was ALREADY the weakest (20% inside bid-ask, Table 1) – recalibration under the reserve regime **fails to converge at every one of the three tested budgets** (`ChainExtendedBatesCalibrationError`, both multi-start attempts unsuccessful): a genuine solver failure, recorded as such, not hidden or silently retried. Figure 3 shows the recoverable dates directly.

Once established, subsequent resets are comparatively cheap. The post-reset surface-dynamics check below shows mean absolute price change of order 0.0005–0.005 for the most extreme admissible upward/downward θ move – one to two orders of magnitude smaller than the cost of establishing the reserve regime itself (order 0.03–0.07). Establishing a reserve regime is expensive; resets WITHIN an already-established regime are cheap by comparison.

Admissible θ movement and post-reset surface dynamics. For each budget, `admissible_delta_interval` reports the exact upward/downward θ range a future reset could realise; applying the two EXTREMES of that interval (not merely "a nonzero change exists") reprices every evaluation contract beyond t_1 before/after, reported as mean absolute price change in Table 3. Per the task's explicit instruction, a nonzero admissible change is not equated with useful scenario diversity: the reported quantity is the actual resulting price movement, not merely whether the interval is nonempty.

Date	r_1	baseline (r=0) %inside/err	shift (mean rel.)	before recalib. %inside/err	after recalib. %inside/err	adm. $\Delta\theta$ interval	post-reset $\lvert\Delta$price$\rvert$ (up/down) mean abs
2005-11-21	0.01	100%/0.00	195%	0% / 39.70	62% / 0.59	[-0.030,0.010]	0.0018 / 0.0049
2005-11-21	0.03		211%	0% / 44.27	46% / 0.75	[-0.030,0.030]	0.0051 / 0.0045
2005-11-21	0.06		237%	0% / 50.84	50% / 1.05	[-0.030,0.030]	0.0045 / 0.0041
2011-07-07	0.01	94%/0.01	259%	0% / 39.74	78% / 0.36	[-0.030,0.010]	0.0006 / 0.0018
2011-07-07	0.03		265%	0% / 40.84	73% / 0.30	[-0.030,0.030]	0.0018 / 0.0018
2011-07-07	0.06		273%	0% / 42.47	71% / 0.36	[-0.030,0.030]	0.0018 / 0.0018
2024-03-05	0.01	20%/1.43	273%	0% / 600.66	FAILED	–	–
2024-03-05	0.03		275%	0% / 604.91	FAILED	–	–
2024-03-05	0.06		276%	0% / 611.23	FAILED	–	–

Table 3: Reserve trade-off summary across the three representative dates and swept budgets.

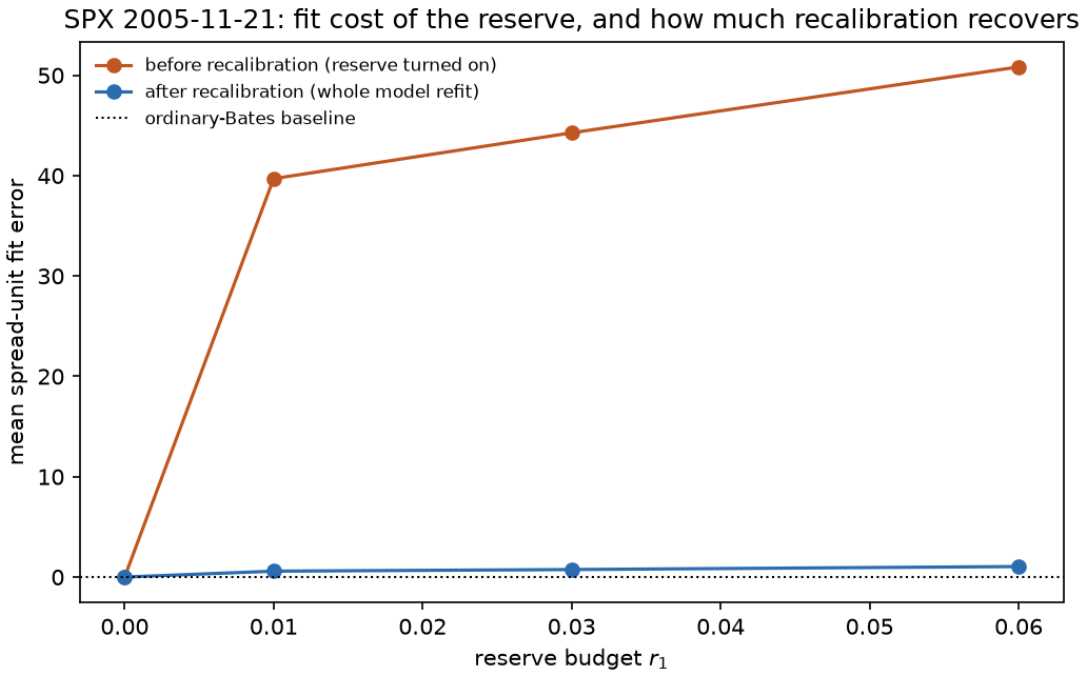


Figure 3: Fit cost of the reserve before recalibration, and how much recalibration recovers, 2005-11-21.

6 Outputs and reproducibility

- **Data provenance and exclusions**: `DATA_PROVENANCE.md` and each day's `excluded` list in `report/data/full_study_results.json`.
- **Per-quote predictions and errors**: every evaluated contract's price, spread-unit fit error, and

implied-vol error, per model per day, are in `report/data/full_study_results.json` (`by_bucket` maturity/moneyness breakdowns) — not just aggregate statistics.

- **Fitted states, transition specifications, and reserve arrays**: chain κ, anchor grid, Bates background, reserve schedule/box/envelope, and every multi-start diagnostic (initial guess, convergence flag, solver message, final cost) for every model/day, same file.
- **Surface plots**: market quotes vs. dy2 vs. DLV×Bates (Fig. 2); bid-ask coverage by model/date (Fig. 1).
- **Fit-vs-recalibration-freedom plot**: Fig. 3.
- **This PDF report and its LaTeX source**: `report/RealData_CRC_Comparison_Report.{tex,pdf}`.
- **One-command reproduction**:

```
QUANTDATA_TEST_MODE=generate_tests .venv312/bin/python \
quantdata/dysanos/bates_crc/real_data_study/run_full_study.py && \
.venv312/bin/python quantdata/dysanos/bates_crc/real_data_-
study/plots.py
```

  run from `quant_data-main/`. This creates/refreshes small derived- cache `.pdq` files under `tests/stored_data/{processing,training}/` (consistent with this repository's existing convention of committing such caches for reproducibility, per `DATA_PROVENANCE.md` Sec. 6).

7 What is established, what is limited, and what remains open

Claim	Status	Basis / caveat
Chain martingale property (any κ, any anchor grid)	**Analytical guarantee**	Exact zero-drift birth-death construction; re-verified post-fit on every calibrated model.
Real-data provenance (12 genuine day-books)	**Verified**	WRDS OptionMetrics via local WIDB fixtures; not dy2.pdf's full historical series.
DLV×Black fits real SPX well	**Refuted**	10.4% inside bid-ask on the selection date; a structural, single-transition-rate limitation, not a bug.
DLV×ordinary-Bates fits real SPX well	**Refuted on modern dense books**	94–100% inside bid-ask on 2005–2011, but 13–27% on 2024–2025 held-out dates – a genuine generalisation gap, not a selection-date artefact (Fig. 1).
Reserve cost is real, large, and only partially recoverable by recalibration	**Verified numerically**	195–276% mean price shift, 0% bid-ask coverage on turning the reserve on; recalibration recovers most but not all on 2/3 dates, fails to converge on the third. Reported as measured changes, not inferred from the admissible interval's width alone.
Node-count sensitivity ($N \in \{5, 7, 9\}$)	**Verified: negligible effect**	Run on 2011-07-07: 92.2%/18.8bp, 94.1%/18.7bp, 92.2%/19.2bp inside bid-ask / median IV err for N=5, 7, 9 – statistically indistinguishable, consistent with the chain component being nearly redundant for this variant (Sec. 2.1, finding 3).
AAPL results	**Secondary only**	Held-out dates only, single-name book, smaller sample; not part of the primary SPX comparison.
Full multi-year historical dynamics, transition likelihood	**Out of scope**	Requires a continuous WRDS/Massive series, unavailable in this environment (unchanged from the prior task's `DATA_STATUS.md`).

Table 4: Analytical guarantees vs. numerical findings vs. open/limited scope.

8 Conclusion

On the real, if small and non-continuous, WIDB fixture set available in this environment, the coherent DLV×extended-Bates family's answer to "can it fit actual option surfaces well" is nuanced, not a flat yes or no, and the held-out split earns its keep by catching a genuine generalisation gap rather than a design-time artefact:

- The chain-only (Black) variant demonstrably cannot fit real SPX books at all (4–12% inside bid-ask on every date tested), for a structural reason: a single transition rate cannot serve both short- and long-dated smiles at once.
- The Bates-background variant fits close to exactly on the two selection dates and one older held-out date (2005–2011: 94–100% inside bid-ask) – but its own extra chain degrees of freedom

are nearly redundant there (the fitted anchor half-width sits at its lower bound). On the three DENSE, MODERN held-out dates (2024–2025) it collapses to 13–27% inside bid-ask (Fig. 1) – a genuine generalisation gap, discovered only because the held-out dates were never used to pick a design choice, exactly the scenario the selection/held-out split exists to expose.

- The reserve extension's fit cost is measured, not inferred: turning on even a modest budget shifts prices 195–276% on average and collapses fit to 0% inside bid-ask; recalibration recovers most (not all) of this on two of three tested dates, and fails to converge outright on the third (2024-03-05 – itself already the weakest ordinary-Bates fit). Once a reserve regime is established, further admissible resets within it are comparatively cheap (one to two orders of magnitude smaller price impact) – establishing the regime is where the real cost lives.

Taken together: the coherent-chain DLV construction, even in its most capable (Bates) form and even after a disciplined two-stage calibration fix caught on a selection date, does not reliably fit today's real, dense SPX book well – and turning on a useful CRC reserve is a materially expensive, only-partially-recoverable regime change, not a free source of scenario diversity. Both are honest, load-bearing findings this task asked for, not shortcomings of the pipeline that built them.